Thow Min Jerald Cham

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ACADEMIC APPOINTMENTS

Nanyang Assistant Professor 2026-Present
School of Electrical and Electronic Engineering, Nanyang Technological University

David and Ellen Lee Postdoctoral Scholar 2024-2026
Division of Physics, Math and Astronomy, California Institute of Technology

EDUCATION

Cornell University, Ph.D., Physics 2018-2024
Thesis Topic: Spin dynamics in antiferromagnetic van der Waals Heterostructures
Advisor: Prof. Daniel C. Ralph
Committee: Prof. Kin Fai Mak, Prof. Erich J. Mueller

Cornell University, M.S. Physics 2018-2021

Imperial College London, Physics BSc, 1st class honors 2014-2017
Research Advisor: Prof. John Tisch

AWARDS & FELLOWSHIPS

David and Ellen Lee Prize Postdoctoral Fellowship, California Institute of Technology 2024-2026

American Physical Society DMP Ovshinsky Award 2024

American Physical Society GMAG Student Travel Award 2024

A*STAR National Science Scholarship Roll of Honor 2017

Imperial College Ken Allen Prize (Top 5 students in Physics) 2017

Imperial College G Research Prize (Top 10 students in Physics, Computing and Math) 2015

Imperial College Physics Department Dean's List 2015, 2017

A*STAR National Science Scholarship 2014-2023

RESEARCH INTERESTS

Antiferromagnetic magnonics and spintronics
Generation, detection, and electrical or optical control of antiferromagnetic magnons and spin dynamics in quantum materials and van der Waals heterostructures.

Ultrafast probing of symmetry-guided quantum coupling pathways
Ultrafast optical spectroscopy of coupled spin, lattice, charge, and orbital degrees of freedom, including light-driven control of collective excitations and transient states.

Nanoscale and terahertz spectroscopy
Development and application of nanoscale optical and terahertz techniques to probe high-frequency excitations in thin crystals, heterostructures, and quantum devices.

CONFERENCE PRESENTATIONS

Invited Talk '*Spin-filter tunneling detection of antiferromagnetic resonance with electrically-tunable damping*', American Physical Society (APS) March Meeting, Denver CO — 2026

Invited Talk '*Spin-filter tunneling detection of antiferromagnetic resonance with electrically-tunable damping*', IEEE Annual Conference on Magnetism and Magnetic Materials, Palm Beach FL — 2025

Invited Talk '*Exploring new spin torque phenomena in van der Waals topological and magnetic heterostructures*', SPIE Optics + Photonics, San Diego CA — 2025

Invited Talk '*Spin-orbit torque in magnetic heterostructures: exchange interactions and ultrasensitive Sagnac magneto-optic interferometry'*, Symposium on Low-Dimensional Magnetic Quantum Materials, Materials Research Society Spring Meeting & Exhibit, Seattle WA — 2024

Focus Session Invited Talk '*Spin dynamics and exchange interactions from a van der Waals antiferromagnet'*, American Physical Society (APS) March Meeting, Minneapolis MN — 2024

Invited Talk *'Exchange bias between magnetic van der Waals materials'*, Cornell NanoScale Science and Technology Facility Annual Meeting, Ithaca NY — 2023

Poster Presentation *'Exchange bias between van der Waals materials: tilted magnetic states and field-free spin-orbit-torque switching'*, Spin Dynamics in Nanostructures Gordon Research Conference, Les Diablerets, Switzerland — 2023

Symposium Invited Talk *'Spin manipulation in van der Waals ferromagnets and antiferromagnets'*, Symposium on Magnetic Manipulation and Excitations in 2D Magnets, APS March Meeting, Las Vegas NV (on behalf of Prof. Dan Ralph). — 2023

Contributed Talk *'Anisotropic gigahertz frequency antiferromagnetic resonance in easy-axis van der Waals semiconductor'*, APS March Meeting, Las Vegas NV — 2023

Poster Presentation *'Anisotropic gigahertz frequency antiferromagnetic resonance in layered van der Waals semiconductor'*, Two Dimensional Electronics Beyond Graphene, Gordon Research Conference, Manchester NH — 2022

Contributed Talk**:** *'Anisotropic gigahertz frequency antiferromagnetic resonance in layered van der Waals semiconductor'*, APS March Meeting, Chicago IL — 2022

Contributed Talk**,** '*Spin-orbit torques generated by low-symmetry $ZrTe_3$*', APS March Meeting — 2021

REFEREED PUBLICATIONS

9 **Cham, T. M. J.**, Chica, D. G., Huang, X., K. Watanabe, T. Taniguchi, Roy, X., Luo, Y. K. and Ralph, D. C., "Spin-filter tunneling detection of antiferromagnetic resonance with electrically-tunable damping." *Science* 389*,* 479-482 (2025).

8 Huang, X., Song, Q., Gurung, G., Pharis, D. A., **Cham, T. M. J.**, Chen, Y., Jain, R., Olszewski, M., Feng, Y., El-Ghazaly, A., Tsymbal, E., Schlom, D. G., and & Ralph, D. C. "Turn-on of current-induced spin torque upon noncollinear antiferromagnetic ordering in delafossite $PdCrO_2$." Advanced Materials, e02670 (2025)

7 Zhou, W., Bishop, A.J., Zhang, X.S., Robinson, K., Lyalin, I., Li, Z., Bailey-Crandell, R., **Cham, T. M. J.**, Cheng, S., Luo, Y.K., Ralph, D.C., Muller, D. A., and Kawakami, R. K., "Tuning the Curie temperature of a 2D magnet/topological insulator heterostructure to above room temperature by epitaxial growth." *Physical Review Materials*, 7(10), 104004-104011 (2023).

6 **Cham, T. M. J.***, Dorrian, R., Zhang, X. S., Dismukes, A. H., Chica, D. G., Roy, X., May A.F., Ralph, D. C., & Luo, Y. K.*, "Exchange bias between van der Waals materials: tilted magnetic states and field-free spin-orbit-torque switching." *Advanced Materials*, 2305739 (2023).

5 Karimeddiny, S.*, **Cham, T. M. J.***, Smedley, O., Ralph, D.C. and Luo, Y.K.*, "Sagnac interferometry for high sensitivity optical measurements of spin-orbit torque." *Science Advances,* 9(36), eadi9039 (2023). First version: arXiv: 2109.13759.

4 **Cham, T. M. J.***, Karimeddiny, Zhang X. S., Dismukes, A. H., Roy, X., Ralph, D. C., & Luo, Y. K.*, "Anisotropic gigahertz antiferromagnetic resonances of the easy-axis van der Waals antiferromagnet CrSBr." *Nano Letters*, *22*(16), 6716-6723 (2022).

3 **Cham, T. M. J.**, Karimeddiny, S., Gupta, V., Mittelstaedt, J. A., & Ralph, D. C. "Separation of artifacts from spin-torque ferromagnetic resonance measurements of spin-orbit torque for the low-symmetry van der Waals semi-metal $ZrTe_3$." *Advanced Quantum Technologies*, *5*(2), 2100111 (2022).

2 Gupta, V., **Cham, T. M. J.**, Stiehl, G. M., Bose, A., Mittelstaedt, J. A., Kang, K., Jiang, S., Mak, K.F., Shan, J. & Ralph, D. C. "Manipulation of the van der Waals magnet $Cr_2Ge_2Te_6$ by spin–orbit torques." *Nano Letters*, *20*(10), 7482-7488 (2020).

1 Tan, T. T., **Cham, T. M. J.**, Reithofer, M. R., Hor, T. A., & Chin, J. M. "Motorized Janus metal organic framework crystals." *Chemical Communications*, *50*(96), 15175-15178 (2014).

*Equal contribution

MANUSCRIPTS IN PREPARATION

3 Smedley, O., **Cham, T. M. J.**, Zheng, X. Y., Channa, S., Riddiford, L., Suzuki, Y., Luo, Y. K. & Ralph, D. C. "Sagnac Interferometer Measurement of Spin-Orbit-Torque Efficiency Acting on the Insulating Magnet $Li_{0.5}(Al,Fe)_{2.5}O_4$." (2026)

2 **Cham, T. M. J.,** Zhao, M.*, Zhou, W.*, Koerner, A., Le, D., Li, Z., Powalla, L., Bergner, D., Thareja, E., Alam, S., Wintz, S., Weigand, M., Hwang, J., Gayles, J., Kawakami, R. K., Luo, Y.K., "Stabilizing bubble magnetic domains in epitaxial 2D magnet/topological insulator heterostructures through interfacial engineering." (2026)

1 Huang, X., Pharis, D. A., Zhou, H., Tian, Z., **Cham, T. M. J.**, Lee, K., Li, Y., Wang, C., Liang, Y., Olszewski, M., Yi, D., Eom, C., Schlom, D. G., Lane, W. M., Shao, D. & Ralph, D. C. "Control of out-of-plane anti-damping spin torque with a canted ferromagnetic spin source." (2026)

ACADEMIC SERVICE

Reviewer for Nature Communications, Newton, AIP Advances, Journal of Applied Physics, Applied Physics Letters, ACS Nano, Physical Review B.

Session chair for Moore Foundation Postdoctoral Symposium (2025), APS March Meeting (2023).

TEACHING, MENTORING & OUTREACH

Teaching Assistant/Grader for Cornell Physics 2216 Special Relativity course.	2019
Singapore National Science Challenge 2017 organizing committee member and judge.	2017
Outreach to Windsor Central High School New York, with Cornell Center for Materials Research (CCMR). Guided students through "Build your own speaker" and "Red cabbage pH meter" lab activity.	2023
Outreach to Pasadena High School students with Caltech. Gave "Introduction to two-magnon scattering in Raman spectroscopy" lecture and lab tours.	2025